THE WEIGHT OF THE BRAIN, OF ITS PARTS AND THE WEIGHT AND LENGTH OF THE SPINAL CORD IN THE RABBIT (RACE X)

HOMER B. LATIMER
Department of Anatomy, University of Kansas
PAUL B. SAWIN[1]
Roscoe B. Jackson Memorial Laboratory

The quantitative study of the brain is not a new subject of investigation, for over a century ago (1836) Tiedeman was studying the weights of the human brain in terms of body weight. Marshall (1892) says that Parchappe in 1836 was the first to relate brain weight to stature. Marshall (1892) and Anthony ('28) have reviewed much of this earlier literature. Snell (1891) and Dubois (1897) are two of the earlier workers who attempted to describe brain weight of man and of other vertebrates by a mathematical formula based on body weight. More recently this series of papers on the relationship of brain weight to body weight has been continued in the studies of von Bonin ('37), Quiring ('41), Count ('48), Sholl ('48) and Wirz ('50), to mention only a few who have contributed to this long and interesting study. The more recent papers have attempted to present formulae for restricted groups of animals rather than for all vertebrates.

The parts of the brain have been studied by Marshall (1892) who studied the cerebrum, cerebellum and medulla of the human brain, and by many others, working first on the

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human brain and later on various vertebrate brains. A few of those who have studied the weights of the major divisions of the brain are: Donaldson ('09 and '24), Dunn ('21), Jenkins ('21), Sutter ('43), Wirz ('50) and Riese and Riese ('52). Larsell and von Berthelsdorf ('41) investigated the masses of parts of the cerebellum with reference to certain muscle masses. Unfortunately all of these workers have not divided the brain in the same manner. A series of animal brains all separated by the same method has been published (Latimer, '24, '25, '38a, '38b, '42, '46, '50 and Corder and Latimer, '49).

Much less has been done on the weights and lengths of the spinal cord. Some of the quantitative studies on the human cord include: Dunn ('21), Lassek and Rasmussen ('39), and Grenell and Scammon ('43). Hines and Emerson ('51) have studied the length of the spinal cord in the infant and adult Macaque. Most of the papers listed above by Latimer have included the ponderal and linear measurements of the cord.

There are few data on the weight of the entire rabbit brain. Crile and Quiring ('40) give the mean brain weight for two male and 22 female Flemish Giant rabbits, Levine et al. ('41) give the percentage weight of the entire brain in 10 adult male New Zealand rabbits, and there are several formulae for estimating the weight of the brain from body weight for the rodent or for the rabbit, a few of which will be discussed. Other aspects of the rabbit brain have been reported, such as the number of cellular elements of the olfactory system (Allison and Warwick, '49), the topography of the rabbit cerebellum (Brodal, '40) and of course other studies of pathways and nuclei. No data have been found in the literature on the weights of the parts of the rabbit brain.

MATERIAL AND METHODS

The 50 male and 50 female rabbits of inbred race X, descended from Castle's small size race, were shipped from the Roscoe B. Jackson Memorial Laboratory in Bar Harbor,

Maine, to Kansas and allowed to rest and regain their shipping weight before they were sacrificed with chloroform. The details of the dissection together with the quantitative study of the other organs is being published elsewhere and will not be repeated here. Three females were in very early pregnancy. The head was separated from the body by an incision passing immediately posterior to the occipital bone and between the occipital condyles and the atlas, thus severing the brain from the cord at the level of the upper root fila of the first cervical nerve. The top of the skull was removed and the brain lifted out. All cranial nerves and the stalk of the hypophysis were cut close to the brain. The meninges and all large blood vessels were removed, leaving only the pia mater to be weighed with the brain, which was divided into 4 parts: prosencephalon, mesencephalon, cerebellum and medulla (including the pons). The method used was the same as in the earlier studies on the cat and dog brains (Latimer, '38a). Each of these parts was immediately placed in a moist chamber until weighed. The total brain weight is the sum of the weights of these 4 parts.

After a complete laminectomy, the dura mater was slit lengthwise, the edges folded back and while still in place in the vertebral canal, the length of the cord from the anterior cut end to the tip of the conus medularis was measured with an inelastic string. This string was used since it would follow the contour of the cord better than the steel tape from which the length of the string was determined. The cord, beginning at the caudal end, was then removed from the vertebral canal, leaving the dura, spinal nerves, severed close to the cord, and all large blood vessels. It was likewise stored in a moist chamber until weighed.

All 4 divisions of the brain and the cord were weighed in weighing bottles on an analytical balance sensitive to 0.1 mg. The parts were weighed in the weighing bottles and other precautions taken throughout the dissection to prevent drying of the parts. All weights were made in the fresh condition and as soon as possible after removal from the animal.

The usual statistical constants were employed in the study of these parts of the central nervous system. The empirical formulae in table 3 were derived by the method of least squares (Snedecor, '38). The weights of the brain and of its 4 parts and the weight and length of the spinal cord were all plotted on body weight and on body length (nose-anus), and these relationships were all linear. When plotted on age there were no observable trends, only the individual scatter of the cases, thus indicating that the youngest were fully mature as to size and that there was no quantitative senescence in the older specimens. These weights and linear dimensions were grouped in 200 gm or 20 mm intervals and these averages used for finding the empirical formulae and also for determining the percentage differences between the calculated and the observed values. These differences are given in parentheses in table 3 following each formula.

There were 15 male and 20 female rabbits carrying the dwarf gene (Greene, '40) which is evidenced in a smaller body weight (Crary and Sawin, '49) and proven by mating. Since the dwarf gene is lethal when homozygous, there are two classes in this study: homozygous non-dwarfs or normals and heterozygotes or dwarfs carrying the dwarf gene. For simplification these will be referred to as normals and dwarfs. The measurements of the parts of the central nervous system of all 50 males and 50 females are given in the first three tables and in tables 4 and 5 similar data are given for these rabbits separated in two groups depending on the presence, or absence, of this dwarf gene.

The gross body weight is the weight of the live animal before anesthetization and the net body weight is the gross body weight minus the weight of the contents of the digestive tube and all fat which could be dissected away, including both subdermal fat and fat from the body cavities.

DIMENSIONS OF THE BODY

Panel A of table 1 shows the means, standard deviations, coefficients of variation and the "t" values of the sex dif-

ferences for the gross and net body weights and the nose-anus length of the 50 male and 50 female rabbits. These data show that the females are significantly heavier and longer than the males, for the females average 17% greater in gross body weight and 5% greater in nose-anus length. The females

TABLE 1

Weight and length of the body for the entire group of 50 males and 50 females in panel A. Panels B and C show similar data for the normal and dwarf males and females

Panel A All rabbits together

	50 MALES			50 FEMALES			"t" VALUES
	Average	Stand. dev.	Coef. var.	Average	Stand. dev.	Coef. var.	
Body weight, gross	1870.0	301.8	16.14	2192.9	459.7	20.96	4.11
Body weight, net	1565.4	244.4	15.61	1734.9	337.3	19.44	2.86
Body length, nose–anus	454.6	24.1	5.30	478.3	33.7	7.05	4.00

Panel B Males
35 normal and 15 dwarf males

	35 NORMAL			15 DWARF			
Body weight, gross	1972.3	274.3	13.91	1630.6	234.1	14.36	4.20
Body weight, net	1650.0	214.6	13.01	1367.9	196.4	14.36	4.32
Body length, nose–anus	463.7	20.8	4.49	433.6	19.4	4.47	4.76

Panel C Females
30 normal and 20 dwarf females

	30 NORMAL			20 DWARF			
Body weight, gross	2480.0	345.4	13.93	1762.1	219.5	12.46	8.24
Body weight, net	1959.3	230.7	11.78	1398.4	139.5	9.98	9.73
Body length, nose–anus	496.8	16.3	3.27	444.3	20.4	4.59	10.20

In tables 1, 2 and 6, "t" values of 2.63 and above are significant at 1% and "t" values from 1.98 to 2.62 are significant at 5%.

are also more variable in weight and in length. The coefficient of variation of the gross body weight is 30% greater in the females and the nose-anus length is 33% greater.

Panels B and C give similar data for the normal and dwarf males and females. The gross body weight of the dwarf males is 17% less than that of the normal males, while

the gross body weight of the dwarf females is 29% less, or the dwarf gene affects the females more than the males. The body length is likewise reduced more in the females, for they are 11% shorter compared with a reduction of 7% in the males. The body weight is more variable in the dwarf males but there is little difference in the variability of the body length. The female body weight is more variable in the

TABLE 2

Weights in grams and percentage weights of the brain and of its parts in 50 male and in 50 female rabbits

	MALES			FEMALES			"t" VALUES
	Average	Stand. dev.	Coef. var.	Average	Stand. dev.	Coef. var.	
Brain	9.156	0.537	5.86	9.349	0.717	7.67	1.48
Prosencephalon	6.118	0.424	6.92	6.218	0.528	8.50	1.03
Mesencephalon	0.753	0.060	8.00	0.785	0.064	8.14	2.56
Cerebellum	1.381	0.089	6.48	1.437	0.122	8.49	2.59
Medulla	0.889	0.076	8.60	0.910	0.072	7.94	1.43
Percentages of net body weight							
Brain	0.596	0.077	12.94	0.554	0.082	14.77	2.66
Prosencephalon	0.398	0.053	13.30	0.368	0.055	15.03	2.74
Mesencephalon	0.049	0.008	15.75	0.047	0.008	17.62	1.55
Cerebellum	0.090	0.013	14.25	0.085	0.012	14.46	2.01
Medulla	0.058	0.007	12.42	0.054	0.008	15.10	2.45
Percentages of total brain weight							
Prosencephalon	66.90	1.45	2.17	66.47	1.13	1.71	1.63
Mesencephalon	8.25	0.61	7.37	8.42	0.62	7.38	1.37
Cerebellum	15.13	0.86	5.71	15.37	0.59	3.81	1.61
Medulla	9.72	0.61	6.32	9.74	0.48	4.88	0.18

normal and the body length is more variable in the dwarf group.

Brain

The first line of table 2 gives the weights in grams of the male and female brains. The female brain is slightly, but not significantly, heavier and also more variable as has been shown for the body weight and length. The brain weights

in both sexes are less than half as variable as the gross body weights. The entire brain in both sexes is less variable than any of its parts. The weight of the brain as a percentage of the net body weight, shown in line 6 of this table, is significantly greater in the males. It is interesting to see that the coefficients of variation of these percentage values are about double the cofficients of the brain weights in grams. The brain is one of the organs which is usually more constant in weight and usually less variable than the total body weight. A comparison of the body weights and brain weights in grams in tables 1 and 2, will show that the body weight is several times as variable as the brain weight and so when these rather constant weights of the brain are expressed as percentages of the variable body weights also in grams, the resulting percentage weights are greater than the brain weights in grams and yet not quite as variable as the body weights. The parts of the brain will be discussed in the following section.

The figures in table 2 give average brain weights for animals of average size and to better estimate the brain weight of animals above or below average body weight, empirical formulae are given in table 3 for estimating either male or female brain weight in grams from gross body weight or nose-anus length. The formulae based on body weight in grams are to be used only from 1300 to 2400 gm for the males and from 1400 to 3200 gm for the females. The formulae for body length may be used only from 400 to 500 mm inclusive for the males and from 410 to 530 mm for the females. The percentages in parentheses following each formula give the average percentage deviation of the calculated from the observed values. These rather small average percentage differences indicate that these formulae fit the cases quite well.

Parts of the brain

Lines 2 to 5 in table 2 give the weights in grams of the 4 divisions of the brain for the males and females. The

"t" values in the last column show that the mesencephalon and cerebellum are significantly heavier at the 5% level and the other two are somewhat but not significantly heavier in the females. All except the medulla are more variable in

TABLE 3

Empirical formulae for estimating the weights of the brain and of its parts and the weight and length of the spinal cord from either body weight in grams or body length in millimeters. X represents either the body weight or nose–anus length

A. Formulae based on gross body weight in grams

	MALES				FEMALES			
Brain	= 0.001X	+	7.212	(2.05%)	= 0.0011X	+	6.9119	(2.09%)
Spinal cord weight	= 0.0008X	+	2.435	(1.64%)	= 0.00079X	+	2.5639	(1.64%)
Spinal cord length	= 0.04X	+	202.0	(0.34%)	= 0.026X	+	230.3	(1.69%)
Prosencephalon	= 0.00069X	+	4.781	(2.59%)	= 0.00073X	+	4.59	(2.78%)
Mesencephalon	= 0.000045X	+	0.6648	(2.80%)	= 0.000066X	+	0.643	(2.13%)
Cerebellum	= 0.0001X	+	1.1878	(2.51%)	= 0.0002X	+	0.9967	(1.91%)
Medulla	= 0.00017X	+	0.5691	(1.14%)	= 0.00011X	+	0.6688	(1.34%)

B. Formulae on nose–anus length in millimeters

Brain	= 0.015X	+	2.297	(1.41%)	= 0.016X	+	1.6875	(2.01%)
Spinal cord weight	= 0.011X	—	1.0634	(0.96%)	= 0.013X	—	1.9181	(1.60%)
Spinal cord length	= 0.5X	+	49.0	(0.28%)	= 0.52X	+	40.0	(0.72%)
Prosencephalon	= 0.01X	+	1.551	(1.55%)	= 0.01X	+	1.405	(2.71%)
Mesencephalon	= 0.0007X	+	0.434	(1.80%)	= 0.00091X	+	0.3647	(3.51%)
Cerebellum	= 0.0017X	+	0.604	(1.18%)	= 0.0027X	+	0.1437	(1.12%)
Medulla	= 0.0022X	—	0.1121	(0.76%)	= 0.0016X	+	0.1449	(0.89%)

the females. In the males, the two suprasegmental parts, telencephalon and cerebellum, are less variable than the parts of the brain stem, but just the reverse is true for the females.

Empirical formulae are given in table 3 for estimating the weights of these 4 divisions of the brain from either gross body weight or body (nose-anus) length.

The middle division of table 2 gives the weights of the brain and its 4 divisions as percentages of the net body weight. The net body weight was used for it represents the real live animal better than the gross body weight which includes the contents of the digestive tube and the reserves of fat. As has been pointed out for the total brain weight, the variability of these percentage weights of the parts of the brain is approximately double that of the weight in grams of these parts. The weights of the brain in grams as well as all of its 4 parts are heavier in the females but all are heavier as percentages of the net body weight in the males, and all except the midbrain are significantly heavier, the prosencephalon alone at the 1% level. The forebrain is unique in being but slightly heavier in grams in the females and relatively the heaviest in the males in percentage of body weight. Just the reverse is true of the midbrain. It is very definitely heavier in its absolute weight in the females and the only part not significantly greater in the males in percentage weight. Thus it seems that the midbrain is relatively greater in the females and the forebrain in the males. The brain and all 4 divisions are heavier in the females but form larger percentages of the net body weight in the males.

The last 4 lines of this table show the weights of the 4 divisions of the brain as percentages of the total brain weight. More than two-thirds of the total brain weight consists of prosencephalon; second in relative weight is the cerebellum, forming a little over 15% of the brain weight. The mesencephalon is relatively the smallest part, being slightly less than the medulla. The two suprasegmental parts together form about 82% of the entire brain leaving but about 18%

for the brain stem or the midbrain and medulla. The supra-segmental parts are very slightly, but not significantly, greater in relative size in the males.

There is less sex difference in these proportions of the total brain weight, as none of these percentages are significantly greater in either sex. The prosencephalon is slightly greater in the male brain and the other three parts are relatively larger in the female brain, with practically no difference in the percentages of the medulla. These coefficients of variation are all low indicating that the proportionate weights of the parts of the brain are rather constant. These coefficients of variation average 5.39 for the males and 4.45% for the females. The similar averages of the coefficients for the weights in grams are respectively, 7.50 and 8.27%. The lowest of these coefficients of the percentages of the brain weight is that of the prosencephalon, but it must be remembered that the forebrain forms over two-thirds of the total brain weight and hence this coefficient should be small. These brains appear to be rather uniform in the proportions of these 4 divisions as percentages of total brain weight and there is no significant sex difference, although the forebrain is slightly larger in the males and the other three parts are slightly larger in the females.

Brains of the normal and dwarf rabbits

The weights and percentage weights of the brain and of its parts in the normal and dwarf males are shown in table 4 and similar data for the females, in table 5. This dwarf gene does not seem to affect the brain weight as much as the body weight, for the male brains are but 4% less in the dwarfs compared to a decrease of 17% in body weight. The female brain like the female body weight is affected more than the male but to a lesser degree than the body weight for the brain is but 10% smaller and the body weight is 29% less in the dwarf females.

The brain and all of its parts are actually heavier in the normal female rabbits, as in the entire group. In the dwarf

rabbits there is less sex difference in absolute weight, with the brain, prosencephalon and medulla slightly heavier in the males and the midbrain and cerebellum very slightly heavier in the females. The "t" values in these two tables show that the weights of the parts of the brain are smaller in the dwarf rabbits with the differences greater in the females than in the males and the least difference seems to be in the midbrain and the male medulla. These are reduced about half as much as the two suprasegmental parts.

With one exception (female medulla) the brain and all of its parts are more variable in the dwarf rabbits than in the normal group. While there is some reduction in the variability in these two groups compared to the entire group, the brain and all of its parts in the dwarf males are more variable than in the entire male group. These average 7.17% in the entire male group and 8.15% in the dwarf males.

The second parts of tables 4 and 5 show the brain and all of its 4 divisions as percentages of the net body weight. All of these percentages are greater in the dwarf males and females compared to the similar normal groups. With the exception of the forebrain in the normal males, all are less variable than in the entire group, but still more variable than the weights in grams as has been shown for the entire group. As percentages of the net body weight, the dwarf rabbits both male and female show the greatest reduction in variability, and unlike the entire group, the females are less variable than the males, with the exception of the female midbrain which is more variable in the dwarf rabbits.

The last divisions of these tables show the weights of the 4 parts of the brain as percentages of the total brain weight. The two suprasegmental parts of the brain in both sexes are greater but not significantly greater in the normal rabbits. With two exceptions, these percentages are less variable in the dwarf males and females indicating that, on the whole, the percentages of the parts of the brain with reference to total brain weight of the dwarf rabbits are less variable in

the females. Although the weight in grams of the parts of the brain are on the average more variable in the male and female dwarf rabbits, they are more uniform in their proportions or as percentages of the total brain weight.

TABLE 4

Weights of the brain and of its parts in 35 normal and in 15 dwarf male rabbits

	NORMAL			DWARF			"t" VALUES
	Average	Stand. dev.	Coef. var.	Average	Stand. dev.	Coef. var.	
			Weights in grams				
Brain	9.265	0.459	4.95	8.849	0.610	6.90	2.63
Prosencephalon	6.241	0.342	5.48	5.898	0.442	7.50	2.95
Mesencephalon	0.751	0.060	7.95	0.756	0.067	8.89	0.27
Cerebellum	1.403	0.079	5.64	1.331	0.096	7.23	2.75
Medulla	0.899	0.075	8.30	0.864	0.077	8.96	1.48
			Percentages of net body weight				
Brain	0.571	0.071	12.36	0.654	0.060	9.20	3.92
Prosencephalon	0.382	0.050	13.03	0.436	0.041	9.40	3.62
Mesencephalon	0.0462	0.0067	14.40	0.0559	0.0058	10.30	4.85
Cerebellum	0.0864	0.0117	13.53	0.0987	0.0097	9.82	3.54
Medulla	0.0551	0.0060	10.87	0.0639	0.0060	9.41	4.71
			Percentages of brain weight				
Prosencephalon	67.02	1.59	2.38	66.63	1.04	1.57	0.86
Mesencephalon	8.12	0.62	7.67	8.55	0.43	5.04	2.40
Cerebellum	15.16	0.86	5.68	15.06	0.89	5.94	0.37
Medulla	9.70	0.68	7.04	9.76	0.52	5.30	0.30

In tables 4, 5, and 7, "t" values of 2.68 and above are significant at 1% and those between 2.01 and 2.67, are significant at 5% level.

In general, the presence of this dwarf gene affects the brain less than the body weight and results in a larger brain weight in grams in the normal group but the brain and all of its parts are larger as percentages of the net body weight in the dwarf males and females. The two parts of the brain stem are larger with reference to brain weight in the dwarf rabbits and the two suprasegmental parts are relatively larger

in the normal groups. These differences in the relative size of the forebrain and cerebellum are not large enough for any definite conclusions, however.

The formulae given in table 3 are valid for these two groups. The formulae are based on body weight or body

TABLE 5

Weights of the brain and of its parts in 30 normal and in 20 dwarf female rabbits

	NORMAL			DWARF			"t" VALUES
	Average	Stand. dev.	Coef. var.	Average	Stand. dev.	Coef. var.	
			Weights in grams				
Brain	9.755	0.493	5.05	8.740	0.577	6.60	6.66
Prosencephalon	6.503	0.393	6.04	5.789	0.411	7.10	6.12
Mesencephalon	0.804	0.057	7.08	0.757	0.065	8.58	2.70
Cerebellum	1.503	0.087	5.79	1.338	0.100	7.45	4.35
Medulla	0.945	0.061	6.44	0.857	0.054	6.30	5.20
			Percentages of net body weight				
Brain	0.504	0.057	11.29	0.629	0.052	8.24	7.80
Prosencephalon	0.336	0.043	12.69	0.416	0.034	8.13	6.95
Mesencephalon	0.0414	0.0041	9.94	0.0545	0.0065	11.95	8.67
Cerebellum	0.0775	0.0072	9.32	0.0962	0.0088	9.17	8.14
Medulla	0.0487	0.0050	10.32	0.0617	0.0053	8.66	8.56
			Percentages of brain weight				
Prosencephalon	66.64	1.29	1.93	66.22	0.81	1.22	1.28
Mesencephalon	8.25	0.54	6.49	8.67	0.67	7.74	2.43
Cerebellum	15.41	0.63	4.12	15.30	0.48	3.11	0.65
Medulla	9.70	0.56	5.74	9.81	0.31	3.15	0.79

length and the dwarf condition affects both body weight and body length as well as brain weight. When these data were plotted on either body weight or body length all of the cases were fairly well distributed around the regression lines, with the smaller cases toward the first part of the line and the larger toward the upper end of the line.

Spinal cord

Table 6 shows the data for the weights and linear measurements of the spinal cord in the 50 male and 50 female rabbits. The cord is significantly heavier, longer and more variable in the females. These weights and lengths as percentages of the net body weight and nose-anus length, respectively, are much alike in the two sexes although both are very slightly greater in the males. The weights of the cord as percentages of the body weight are 24% more variable in the males and 17% more variable in the females than the corresponding weights in grams. There is little sex difference in the

TABLE 6

Weights, lengths and percentage weights and lengths of the spinal cord in 50 male and 50 female rabbits

	50 MALES			50 FEMALES			"t" VALUE
	Average	Stand. dev.	Coef. var.	Average	Stand. dev.	Coef. var.	
Cord, gm	3.925	0.384	9.78	4.292	0.470	10.96	4.23
Cord, mm	276.5	13.8	4.98	288.3	16.0	5.56	3.64
Cord, % of net B.W.	0.255	0.031	12.15	0.253	0.032	12.85	0.41
Cord L., % of B.L.	60.9	1.69	2.77	60.7	1.75	2.88	0.58

variability of the linear dimensions of the cord as percentages of the body length. These coefficients of variation of the cord length as percentages of nose-anus length are very low. These two percentages and that of the prosencephalon as a percentage of the total brain weight are the lowest of all of the coefficients of variation. These linear dimensions naturally should be less, or approximately one-third those of the three dimensional measurements of volume or weight, and as has been stated, the prosencephalon forms over two-thirds the weight of the entire brain and hence this coefficient would be expected to be small.

Table 7 shows similar data for the spinal cord of the normal and dwarf males and females. The empirical formulae given

in table 3 may be used for the dimensions of all of the spinal cords, including the cords of the normal and dwarf rabbits.

The weight and length of the cord is greater in both sexes in the normal rabbits and all but the weight of the cord in the males are significantly greater as shown by the "t" values in the last column. The average weight of the

TABLE 7

Weights, lengths and percentage weights and lengths of the spinal cord in the normal and dwarf rabbits

	NORMAL			DWARF			"t" VALUES
	Average	Stand. dev.	Coef. var.	Average	Stand. dev.	Coef. var.	
35 normal and 15 dwarf males							
Cord, gm	3.945	0.365	9.26	3.915	0.438	11.19	0.24
Cord, mm	280.6	12.1	4.32	267.0	13.0	4.89	3.62
Cord, % of net B.W.	0.241	0.022	9.19	0.288	0.023	8.08	6.70
Cord L., % of B.L.	60.5	1.49	2.47	61.6	1.94	3.16	2.16
30 normal and 20 dwarf females							
Cord, gm	4.537	0.38	8.31	3.926	0.359	9.15	5.72
Cord, mm	298.5	9.6	3.21	273.0	12.3	4.49	8.38
Cord, % of net B.W.	0.233	0.022	9.21	0.282	0.024	8.43	7.40
Cord L., % of B.L.	60.1	1.62	2.70	61.5	1.64	2.68	2.94

cord in the dwarf males is 0.25% less than in the entire group and the dwarf female cord is 8.5% lighter in weight. Similar reductions in the length of the cord are: males, 3.4% and females, 5.3%. Thus the weight and length of the cord are reduced by the dwarf gene and again the females are affected more than the males as has been shown for the dimensions of the body and of the brain. The weights and linear measurements are less variable in the normal but the percentages are slightly more variable.

Tables 6 and 7 both show that the spinal cord as well as the brain is heavier and also longer in the females, and in the normal compared to the dwarf rabbits. The percentages, however, show that the dimensions of the cord follow the body weight and length more closely than do the brain measurements. All of these measurements are slightly less variable than the similar measurements of the entire group shown in table 6, except the cord weight in the dwarf males.

Correlations

Table 8 shows the correlations between the two dimensions of the body and the seven measurements of the central nervous system, with those for the males shown to the left and lower part of the table, those for the females above and to the right of the diagonal row of dashes. The averages along the bottom of the table show the average of all male correlations with the part shown at top of column. Similar averages for the females are shown in the last column with the part listed in the first column. All of these correlations are positive and so all signs are omitted. All correlations are significant at the 1% level except the 5 male correlations marked "*", which are significant at the 5% level.

The average of all of the male correlations is 0.550 and for the females, 0.686, or the correlations average 25% greater in the females, although the averages of the coefficients of variation in tables 1 and 2 are greater in the females. The female brains vary more in size but they retain their relative proportions better than the males. The range of these correlations is from 0.286, between male body length and the midbrain, to two correlations above 0.9, or 0.979 between the female brain and forebrain and a correlation of 0.945 between the male body weight and body length. There are 4 correlations in the males above 0.8, and 6 in the females. There are 15 correlations below 0.5 in the males, 7 of which are correlations with the midbrain. In the females there are but 6 correlations below 0.5 and of these 5 are with the

TABLE 8

All correlations are positive and significant at the 1% level except those marked "" which are significant at the 5% level. Male correlations are in lower left half and female correlations are in upper right. The averages are for all correlations within the same sex and for the part indicated.*

	BODY WEIGHT	BODY LENGTH	BRAIN	CORD WEIGHT	CORD LENGTH	PROS.	MESEN.	CEREB.	MEDULLA	AVERAGE OF ALL FEMALE CORRELATIONS
Body W.	...	.764	.718	.789	.807	.643	.472	.743	.720	.707
Body L.	.945	...	.641	.725	.845	.592	.482	.583	.622	.657
Brain	.657	.587	...	.766	.660	.979	.604	.877	.790	.754
Cord W.	.674	.645	.519	...	.808	.641	.709	.752	.889	.760
Cord L.	.871	.859	.565	.660	...	.488	.435	.700	.678	.678
Prosen.	.562	.512	.894	.342	.484	...	.429	.791	.688	.656
Mesen.	.323*	.286*	.449	.574	.316*	.320*	...	.489	.634	.532
Cerebellum	.478	.394	.549	.485	.430	.386	.324*	...	.735	.709
Medulla	.661	.660	.662	.737	.532	.489	.435	.520	...	.720
Av. of all male correlations	.646	.611	.610	.576	.590	.499	.378	.446	.587	

mesencephalon. The only correlations above 0.5 with the mesencephalon are with the cord weight in both sexes, female brain weight and weight of the female medulla. Thus the mesencephalon seems to be the least related in weight to the other dimensions.

The correlations between the two suprasegmental parts, prosencephalon and cerebellum, and the two parts of the brain stem are good correlations in the females but not as high in the males. The weights of the brain and the cord have higher correlations with body weight than with body length in both sexes. The cord length is better correlated with body weight in the males but its correlation with body length is higher in the females. Naturally the correlations between the entire brain weight and that of the forebrain are the highest correlations with the weight of the entire brain.

The 4 groups of correlations in this table having the highest averages in both sexes are with: body weight, brain weight, body length and cord weight, but not in the same order in both sexes. The group showing the greatest sex difference is the group of correlations with the prosencephalon; the male correlations average but 0.398 and the females 0.636% or nearly 60% more than the average for the males.

In general, the body weight and length are significantly correlated with each other, with the brain weight, and with the weight and length of the spinal cord. The brain is significantly correlated with all 4 divisions of the brain except with the midbrain in the males.

DISCUSSION

There are but few data on the weight of the entire rabbit brain. Crile and Quiring ('40) give the average brain weight and the average body weight of 22 female and 2 male Flemish Giant rabbits which are heavier than race X. From their data, the average percentage of the female brain weight with reference to body weight was found to be 0.39%, which is less than the percentage of 0.44 of race X females. Both

of these are percentages of the gross body weight. Quiring ('46) states that animals from colder climates have relatively heavier brains. These race X rabbits were raised out of doors in Maine and Quiring's rabbits were from Cleveland, and whether raised out of doors or not is not stated. It is well known that smaller animals have relatively heavier brains, and so whether due to climate or body size, these race X rabbits do have relatively heavier brains than those reported by Crile and Quiring. Levine et al. ('47) find the total brain weight of 10 New Zealand male rabbits averaging 0.32% of the gross body weight, which corresponds to the percentage given by Crile and Quiring. Both of these races are heavier than race X and hence would be expected to have relatively lighter brains. No data on the weights of the parts of the rabbit brain have been found in the literature.

This study is not concerned with proving or disproving the validity of the various cephalization formulae, but three of these were tried on race X, to see how well they would predict the brain weights from the body weights. Using the formula given by Quiring ('41) the average brain weight of the females was but 1.12% below the observed average, but the male predicted average was 7.37% too low. This exponential formula was used to compute the averages for the entire regression line as was done with the formulae given in table 3. The average percentage difference was 7.74% for the males and 11.36% for the females, or over three and 5 times the averages for the straight line formulae from table 3 for the weight of the entire brain.

The formula given by Sholl ('48) for Set I did not give average weights as close to the observed values as Quiring's formula. All of the averages computed from Sholl's formula were from 24 to over 40% above the observed averages for both the males and the females. These formulae gave results too high when used in place of the linear formulae in table 3. The formula of von Bonin ('37) gave averages of 6.28% for the males and 7.79% for the females and in both sexes too high. When the formula of von Bonin was used to estimate

the weights of the individual cases, the average percentage differences were 8.90 for the males and 15.16% for the females. All of these exponential formulae rise too rapidly and most of them are above the individual observed averages.

These exponential formulae are designed to fit the average brain weights of larger groups of animals and the formulae of Quiring and von Bonin do give average weights of the brains of this series of rabbits with an error of less than 10%, but the individual brain weights of this race of rabbits are best fitted by the linear formulae.

The body weight in the rabbit, as in other animals, is more variable than brain weight. The coefficients of variation of the body weights for all of race X together (tables 1 and 2) are 2.7 times the coefficients of variation of the brain weights. In the normals of both sexes this ratio is about 2.8 times. In the dwarf rabbits, both male and female, this ratio decreases to about 2 times, due to the greater variability of the brain weights in both sexes. In the guinea pigs (Latimer, '50) due to less variation in body weight, this ratio decreases to about 1.8 times. In the adult cats (Latimer, '38b) this ratio is three times for the males and 2.7 times for the females. The female body weights in the cat are less variable than in the males. In the adult dogs, this ratio is again about three times. Thus in all of these animals the body weight is from two to three times as variable as the weight of the brain. In all of these animals the brain appears to have developed more or less according to its own pattern and somewhat irrespective of the body size. Marshall (1892), in his study of the size of the human brain, comments on the comparative independence of the brain weight with regard to stature. The figures given by Marshall to show the extent and type of variation in the human brain have been changed to percentages and they are: variations due to sex, 10.67%; due to age, 6.39%; and due to stature, 4.27%. Marshall has even suggested the regressive changes in the brain weight due to age and thus has preceded some of the more modern work on gerontology.

The weights of the parts of the brain relative to the entire brain weight for a series of 8 animals, including man, have been shown as bar graphs in an earlier report (Latimer, '50, fig. 2). These brains with the addition of the rabbit brain are, in order of increasing relative weight of the prosencephalon as a percentage of the entire brain weight: frog, 50%; chicken, 55%; rabbit, female, 66% and male, 67%; rat, 67%; turtle, 68%; guinea pig, 70%; cat, 75%; dog, 83%; and man, 87%. This is not an attempt to place these animals in the order of their increasing intelligence for there are too many problems of structure and organization in the forebrain to arrange these animals entirely on the basis of the weight of the prosencephalon.

The midbrain of the rabbit is about the same relative size as that of the guinea pig and much smaller than in the frog, chicken and turtle, with all of the others having smaller midbrains, relative to the entire brain weight. The rabbit cerebellum is similar in relative size to that of the chicken, rat, and guinea pig. The rat and turtle have relatively larger medullae and in the chicken and guinea pig the medulla is relatively smaller. In general, the size of the midbrain as a percentage of the brain weight decreases in this series of animals, arranged with an increasing size of the prosencephalon. The cerebellum varies but little except in the sluggish turtle. It was not separated from the medulla in the frog, and the medulla of the turtle, like the cerebellum, is large.

The forebrain is slightly but not significantly larger in the male rabbits as in the male cats (Latimer, '38b), but it is slightly larger in the female dogs (Latimer, '42). There is little sex difference in the weight of the prosencephalon in the normal rabbits, but in the dwarf rabbits, the male has a slightly heavier forebrain. The midbrain in the rabbit as in the cat and guinea pig is the most variable part of the brain in terms of percentage of the total brain weight.

Average weights and percentage weights of the cerebellum with reference to the entire human brain weight, computed from the data given by Mall ('09), show a sex difference in

the negro, with the male negro cerebellum significantly heavier. The white male cerebellum is slightly but not significantly heavier than the cerebellum in the negro male. These averages and percentages are given as computed from Mall's data, in an earlier publication (Latimer, '46, table 2).

Larsell and von Berthelsdorf ('41) find that crus I of the cerebellum is larger in animals having large fore limb muscles and crus II is larger in the rat and rabbit, or in animals with large hind limb muscles, and furthermore, the ratios between the sizes of these two crura are about the same as the ratios between the muscle masses of the fore and hind limbs.

Sutter ('43) has studied the postembryonic growth of the parts of the brain in two species of birds, and Wirz ('50), using the same method of dividing the brain, has attempted to determine the "cephalization" of a number of animals. Their method of dividing the brain is not the same as that used in this series of animals. Sutter shows by a diagram the method of removing the cerebellum, corpora bigemina and the upper part of the forebrain from the remaining parts of the brain stem. The line of separation follows, in general, the line of the embryonic sulcus limitans.

Riese and Riese ('52) in a study of 42 baboon brains find the hemispheres forming 87%; cerebellum, 9.5% and the medulla, including the pons, 2.4% of the entire brain weight.

There are no known data on the weights of the parts of the rabbit brain other than the work of Larsell and von Berthelsdorf which has already been discussed. Allison and Warwick ('49) have reported the weights of the olfactory system, and Kimmel ('40), the weights of the motor nuclei in the rabbit brain, but no other weights of the parts of the brain in the rabbit have been found in the literature.

The correlations of the parts of the central nervous system in the rabbit, cat (Latimer, '38b) and the dog (Latimer, '42) are, on the average, higher in the females than in the males. In these rabbits the average of the correlations with body weight is better than with body length, which is just the

reverse of the condition found in the dog. The mesencephalon is poorly correlated with the other parts of the brain in the dog, cat and guinea pig (Latimer, '50) as well as in these rabbits, and in all of these it is exceedingly variable. Some of this variability may be due to the fact that it was divided at both ends from the other parts of the brain, and the separation of the midbrain from the forebrain was the most difficult to make accurately and uniformly. The midbrain is becoming, in the more highly developed animals of this series, very largely just a connection with the parts of the brain anterior and posterior to it. It does not keep pace with the growth of the brain, for it forms 24% of the total weight in the frog and but 2.7% in the dog.

The weight of the spinal cord of the rabbit is 42.9% of the weight of the brain in the males and 45.9% in the females. Comparing these values with the percentages shown in figure 1 (Latimer, '50) we find but three larger percentages of the spinal cord weight, namely: frog, 46%; chicken, 90%; and the turtle with a cord forming 120% of the brain weight. In man the cord weight is but 2% of the brain weight. If one may draw any physiological conclusions from these quantitative relations, one would think that a large part of the control of the muscular activity of the rabbit centers in the cord.

Quantitative studies of the growth of the fetal human spinal cord have been made by Lassek and Rasmussen ('39) and Grenell and Scammon ('43). The adult human cord has been studied by McCotter ('16) and Lassek and Rasmussen ('38). Hines and Emerson ('51) have studied the growth of the cord in the Macaque. No quantitative data have been found for the spinal cord in the rabbit.

The length of the spinal cord with reference to body length in the rabbit shows but little sex difference for it forms 60.7% of the body length in the female and 60.9% in the male. In the male guinea pig the cord is 52.7% of the nose-anus length, in the male dog, 63.4% and in the female dog, 62.5%. McCotter ('16) reports a length of 26.4% in white males and

26.2% in white females. Lassek and Rasmussen ('38) report a cord length in the human newborn of 31.3% and in the adult 24.4%. These percentages for man are with reference to stature rather than stem length and hence much smaller than the percentages against nose-anus length. In the rabbit, dog and man the cord in the females forms a slightly smaller percentage of the body length or of the stature.

SUMMARY

In this series of 50 male and 50 female rabbits, the body weight, nose-anus length, weights of the brain and of its parts and the weight and length of the spinal cord are greater and also more variable in the females. As percentages of the net body weight, the brain and its parts are larger in the males.

Empirical formulae are given for estimating the weight of the brain or any of its 4 parts and the weight and length of the cord, from either gross body weight or body length.

The forebrain forms 67% of the entire brain weight; the mesencephalon, 8%; the cerebellum, 15%; and the medulla, including the pons, 10%. The weights in grams of the 4 parts of the brain are all heavier in the females, but as percentages of the net body weight they are heavier in the males. The forebrain is slightly greater in the males and the midbrain in the females. The parts of the brain as percentages of the entire brain weight are less variable than their weights in grams and also as percentages of the net body weight.

Fifteen males and 20 females of these rabbits carried the dwarf gene. This reduced the body weight of the males an average of 17% and the brain weight, 4.5%. Similar reductions in the females are: body weight, 29% and brain weight, 10%. Thus the females are affected more than the males and the brain weights less than the body weights.

The brain and all of its parts show less sex difference in the dwarf rabbits due to a greater reduction in the weights of the brain and its parts in the females. The greatest reduction appears to be in the weights of the mesencephalon

and medulla. The weights of the parts of the brain as percentages of the total brain weight are most constant and these are more constant in the dwarf than in the normal groups, showing the rather uniform proportions of these brains, especially in the dwarf rabbits.

The cord weight and length are also greater in the females but again the percentages are slightly greater in the males. The weight of the cord in the dwarf males is reduced by 0.25% and in the females by 8.5%. The similar reductions in the cord length are: 3.4% in the males and 5.3% in the females.

Correlations are given for body weight, body length, brain weight, the weights of the 4 divisions of the brain and the weight and length of the spinal cord. These correlations are all positive, but range from 0.286 to 0.979. The mesencephalon has the lowest correlations of any part of the brain. The average of the correlations is greater in the females. Although the female brains are more variable, their proportions are more constant than in the males.

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